

Assignment 2: Coding Basics

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OVERVIEW

This exercise accompanies the lessons/labs in Environmental Data Analytics on coding basics.

Directions

1. Rename this file `<FirstLast>_A02_CodingBasics.Rmd` (replacing `<FirstLast>` with your first and last name).
2. Change “Student Name” on line 3 (above) with your name.
3. Work through the steps, **creating code and output** that fulfill each instruction.
4. Be sure to **answer the questions** in this assignment document.
5. When you have completed the assignment, **Knit** the text and code into a single PDF file.
6. After Knitting, submit the completed exercise (PDF file) to Canvas.

Basics, Part 1

1. Generate a sequence of numbers from one to 55, increasing by fives. Assign this sequence a name.
2. Compute the mean and median of this sequence.
3. Ask R to determine whether the mean is greater than the median.
4. Insert comments in your code to describe what you are doing.

```
#1. Sequence 1 -5, intervals by 5
sequence1 <-seq(1,55,by=5)
# <- assigns a name to the object
# function inputs for seq are starting number, ending number, by is the intervals

#2. Mean & Median
meanseq<-mean(sequence1)
medianseq<-median(sequence1)
#since I named the sequence, I can use the functions mean and median with sequence1 as the input

#3. Which is greater mean or median
if (meanseq > medianseq)'true' else('false')
```

```
## [1] "false"
```

```
#if statement to see if mean is greater than median, 'true' or 'false' outcome dependent on which is gr
```

Basics, Part 2

5. Create three vectors, each with four components, consisting of (a) student names, (b) test scores, and (c) whether they are on scholarship or not (TRUE or FALSE).
6. Label each vector with a comment on what type of vector it is.
7. Combine each of the vectors into a data frame. Assign the data frame an informative name.
8. Label the columns of your data frame with informative titles.

```
#5.  
#creating vectors  
namevector<-c('John', 'Susie', 'Tyrell', 'Kendra') #character vector  
scorevector<-c(60,70,80,90) #numeric vector  
scholarvector<-c(FALSE, TRUE, FALSE, TRUE) #logic vector  
#combining vectors into a dataframe  
studentsinfo <- data.frame(namevector,scorevector,scholarvector)  
#naming columns of dataframe  
names(studentsinfo)<-c('Student Name','Test Score','Scholarship')
```

9. QUESTION: How is this data frame different from a matrix?

Answer:A data frame is different than a matrix because its columns can have different types of data (numbers, characters, logic). Whereas in a matrix all the data must be the same type.

10. Create a function with one input. In this function, use `if...else` to evaluate the value of the input: if it is greater than 50, print the word “Pass”; otherwise print the word “Fail”.
11. Create a second function that does the exact same thing as the previous one but uses `ifelse()` instead of `if...else`.
12. Run both functions using the value 52.5 as the input
13. Run both functions using the **vector** of student test scores you created as the input. (Only one will work properly...)

```
#10. Create a function using if...else  
myfunction<-function(x){  
  if(x>50) 'Pass'  
  else 'Fail'  
}  
  
#11. Create a function using ifelse()  
function2<-function(y){  
  ifelse(y>50, 'Pass', 'Fail')  
}  
  
#12a. Run the first function with the value 52.5  
myfunction(52.5)
```

```
## [1] "Pass"
```

```
#12b. Run the second function with the value 52.5  
function2(52.5)
```

```
## [1] "Pass"
```

```
#13a. Run the first function with the vector of test scores - does not work  
#myfunction(scorevector)
```

```
#13b. Run the second function with the vector of test scores - uses ifelse  
function2(scorevector)
```

```
## [1] "Pass" "Pass" "Pass" "Pass"
```

14. QUESTION: Which option of `if...else` vs. `ifelse` worked? Why? (Hint: search the web for “R vectorization”)

Answer: ‘ifelse’ worked because this function is vectorized meaning it automatically can perform the function on each element within the vector. Whereas the ‘if’...‘else’ function is not vectorized and would require a loop using ‘for’ to perform the function on each element in the vector.

NOTE Before knitting, you’ll need to comment out the call to the function in Q13 that does not work. (A document can’t knit if the code it contains causes an error!)